ONE-RELATOR GROUPS ARE COHERENT

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Abstract. Building on a recent breakthrough of Jaikin-Zapirain, we show that all one-relator groups are coherent.

1. Introduction

A group G is *coherent* if every finitely generated subgroup of G is finitely presented. It is a long-standing conjecture of Baumslag [Bau74] that all one-relator groups are coherent.

One-relator groups are known to be generically coherent by work of Sapir–Špakulová [SŠ11] and Kielak–Kropholler–Wilkes [KKW22]. One-relator groups with torsion are known to be coherent by independent work of Louder–Wilton [LW20] and Wise [Wis22]. Coherence of one-relator groups with negative immersions was established by Louder–Wilton [LW21]. Kielak and the author showed in [KL23] that hyperbolic and virtually compact special one-relator groups are virtually free-by-cyclic and thus coherent by Feighn–Handel [FH99].

A group G is *homologically coherent* if every finitely generated subgroup of G has type FP_2. Jaikin-Zapirain recently achieved the following breakthrough.

Theorem 1.1. [JZ23] *Torsion-free one-relator groups are homologically coherent.*

Finitely presented groups are of type FP_2, but for a while it was an open question as to whether there existed a group of type FP_2 that was not finitely presented. The first examples of such groups are due to Bestvina–Brady [BB97]. It is the purpose of this note to show that for one-relator groups, coherence and homological coherence are equivalent. By Jaikin-Zapirain's result, we confirm Baumslag's conjecture.

Theorem 1.2. *One-relator groups are coherent.*

A consequence of Theorem 2.5 and our proof of Theorem 1.2 is also the fact that coherence and homological coherence are equivalent for any group that can be built up from coherent groups by iteratively performing amalgamated free products and HNN-extensions.

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2. The proof

The reader is referred to Serre's monograph [Ser03] for the necessary background in Bass–Serre theory. We will always assume that our trees and group actions are simplicial. If G is a group acting on a tree $\mathcal{T}$, then we call $\mathcal{T}$ a *G-tree*. If $H \subseteq G$, we denote by $\mathcal{T}/H$ the graph obtained from $\mathcal{T}$ by identifying each point $t \in \mathcal{T}$ with gt for each $g \in H$. A G-tree $\mathcal{T}$ is *cocompact* if $\mathcal{T}/G$ is compact. A map between trees $\mathcal{S} \to \mathcal{T}$ is a *morphism* if it sends vertices to vertices and edges to (possibly trivial) edge paths. A G-tree $\mathcal{T}$ is *dominated* by another G-tree $\mathcal{S}$ if there is a G-equivariant morphism $\mathcal{S} \to \mathcal{T}$. We call such a morphism a *domination map*.

Lemma 2.1. *Let G be a group and let $\mathcal{S}$ be a G-tree. If $N_1, \dots, N_k \leqslant G$ are subgroups that stabilise vertices, then $\mathcal{T} = \mathcal{S}/\langle\langle N_1, \dots, N_k \rangle\rangle$ is a $G/\langle\langle N_1, \dots, N_k \rangle\rangle$-tree.*

Proof. Suppose for a contradiction that $\mathcal{T}$ is not a tree; then there is an embedded loop in $\mathcal{T}$. In particular, there is an element $g \in \langle\langle N_1, \dots, N_k \rangle\rangle$ and an embedded loop in $\mathcal{S}/\{g\}$ that maps injectively to an embedded loop in $\mathcal{T}$. Since conjugates of elements that stabilise vertices also stabilise vertices, $g = g_1 g_2 \dots g_n$ where each g_i stabilises a vertex in $\mathcal{S}$. We have that $\mathcal{S}/\{g\} \to \mathcal{T}$ factors through $\mathcal{S}/\{g_1, \dots, g_n\}$, which hence must also contain an embedded loop. As g_1 stabilises a vertex, we see that $\mathcal{S}/\{g_1\}$ is a tree. Similarly, since g_i stabilises a vertex in $\mathcal{S}/\{g_1, \dots, g_{i-1}\}$ for each i, we see that $\mathcal{S}/\{g_1, \dots, g_n\}$ must also be a tree. But this is a contradiction and so $\mathcal{T}$ must be a tree. $\square$

The following result is a direct consequence of [GL17, Proposition 2.17] due to Guirardel–Levitt, which generalises results of Baumslag–Shalen [BS90, Theorem 1] and Fujiwara–Papasoglu [FP06, Proposition 5.12].

Proposition 2.2. *Let G be a group and let $\mathcal{T}$ be a cocompact G-tree. If G is finitely presented, then there is a cocompact G-tree $\mathcal{S}$ with finitely generated vertex and edge stabilisers that dominates $\mathcal{T}$.*

Before proceeding, we first need the following useful lemma due to Bieri–Strebel [BS78, Lemma 2.1].

Lemma 2.3. *A group G is of type FP_2 if and only if there is a short exact sequence*

$$1 \longrightarrow N \longrightarrow H \longrightarrow G \longrightarrow 1$$

where H is finitely presented and N has trivial abelianisation.

We now generalise Proposition 2.2 to groups of type FP_2. The proof is essentially an extension of the proof of [BS78, Theorem A].

Theorem 2.4. *Let G be a group and let $\mathcal{T}$ be a cocompact G-tree. If G is of type FP_2, then there is a cocompact G-tree $\mathcal{S}$ with finitely generated vertex and edge stabilisers that dominates $\mathcal{T}$.*

Proof. Since G is of type FP_2, there is a short exact sequence

$$1 \longrightarrow N \longrightarrow H \xrightarrow{\pi} G \longrightarrow 1$$

where H is finitely presented and the abelianisation of N is trivial by Lemma 2.3. Now π turns $\mathcal{T}$ into an H-tree. Since H is finitely presented, by Proposition 2.2 there is a cocompact H-tree $\mathcal{S}'$ with finitely generated vertex and edge stabilisers that dominates the H-tree $\mathcal{T}$. Since $\mathcal{S}'$ is cocompact, there are finitely many vertex orbit representative $v_1, \ldots, v_k$ in $\mathcal{S}'$. Denote by $H_1, \ldots, H_k \leqslant H$ the corresponding stabiliser subgroups, which are finitely generated. Let $N_i = \ker(\pi|_{H_i})$. We now consider the quotients

$$Q = H/\langle\langle N_1, \ldots, N_k \rangle\rangle,$$
$$\mathcal{S} = \mathcal{S}'/\langle\langle N_1, \ldots, N_k \rangle\rangle.$$

By Lemma 2.1, $\mathcal{S}$ is a Q-tree. Since $\mathcal{S}/Q \cong \mathcal{S}'/H$, we see that $\mathcal{S}$ is a cocompact Q-tree. As π factors through $H \to Q$, we obtain a domination map of cocompact Q-trees $\mathcal{S} \to \mathcal{T}$. Denote by N_Q the image of N in Q. By definition, N_Q intersects each vertex stabiliser of the Q-tree $\mathcal{S}$ trivially and thus acts freely on $\mathcal{S}$. A group acting freely on a tree is free and so N_Q must be free. Since N had trivial abelianisation and N_Q is a quotient of N, this implies that N_Q also has trivial abelianisation. A free group has trivial abelianisation only if it is trivial and hence $N_Q = 1$ and $Q \cong G$. This makes $\mathcal{S}$ a cocompact G-tree, dominating the G-tree $\mathcal{T}$. The vertex stabilisers of the G-tree $\mathcal{S}$ are conjugates of $H_1/N_1, \ldots, H_k/N_k$ and so are finitely generated. Similarly, the edge stabilisers are conjugates of quotients of the edge stabilisers of the H-tree $\mathcal{S}'$ and so are also finitely generated. This completes the proof. □

Using Theorem 2.4, we obtain a criterion for coherence of a group acting on a tree with coherent vertex stabilisers.

Theorem 2.5. *Let G be a group and let $\mathcal{T}$ be a cocompact G-tree with coherent vertex stabilisers. If G is homologically coherent, then G is coherent.*

Proof. Let $H \leqslant G$ be a finitely generated subgroup. Since H is finitely generated, there is an H-invariant subtree $\mathcal{S} \subset \mathcal{T}$ such that $\mathcal{S}$ is a cocompact H-tree. Now each vertex stabiliser for the action of H on $\mathcal{S}$ is contained in a vertex stabiliser for the action of G on $\mathcal{T}$. Since H has type FP_2, by Theorem 2.4, H splits as a finite graph of groups with

finitely generated vertex and edge groups, each conjugate into vertex stabilisers for the action of G on $\mathcal{T}$. Since the vertex stabilisers for the action of G on $\mathcal{T}$ are coherent, it follows that H splits as a finite graph of groups with finitely presented vertex and edge groups. Hence, H is finitely presented. □

Before proving Theorem 1.2 we need one final ingredient. The following result was first shown by Masters [Mas06, Theorem 4.1] and also appears as [Lin22, Theorem 5.4].

Theorem 2.6. *If G is a finitely generated one-relator group, there is a finite sequence of finitely generated one-relator subgroups*

$$G_N \leqslant \cdots \leqslant G_1 \leqslant G_0 = G$$

such that G_N is finite and for all $i < N$, G_i splits as an HNN-extension over G_{i+1} with finitely generated free associated subgroups.

Proof of Theorem 1.2. Let G be a one-relator group. If G is not finitely generated, it is isomorphic to a free product of a finitely generated one-relator group with a free group. By the Kurosh subgroup theorem and Grushko's theorem, any finitely generated subgroup of G splits as a free product of conjugates of finitely generated subgroups of the free factors of G. Since free groups are coherent, if finitely generated one-relator groups are coherent, all one-relator groups are coherent. Hence, from now on we may assume that G is finitely generated. Since one-relator groups with torsion are coherent by [LW20] and [Wis22], we may further assume that G is torsion-free.

By Theorem 2.6, there is a finite sequence of of finitely generated one-relator subgroups $G_N \leqslant \cdots \leqslant G_1 \leqslant G_0 = G$ such that G_N is trivial and for all $i < N$, G_i splits as an HNN-extension over G_{i+1}. Hence, for each $i < N$, the Basse–Serre tree associated with the corresponding HNN-extension yields a cocompact G_i-tree $\mathcal{T}_i$ whose vertex stabilisers are conjugates of G_{i+1}. If G_{i+1} is coherent, then since G_i is homologically coherent by Theorem 1.1, G_i is also coherent by Theorem 2.5. Since G_N is certainly coherent, an inductive argument completes the proof. □

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